

Harshavardhan Robba

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building production LLM systems—RAG pipelines, real-time voice agents, agentic workflows, and code-generation platforms—using LangChain, LangGraph, Azure OpenAI, and OpenAI Realtime. Designs systems with evaluation and guardrails baked in (DeepEval, CI/CD gating) and instruments them with OpenTelemetry for reliable performance at scale. Strong backend foundation in Node.js/Python with a track record of shipping multi-tenant, API-first services. U.S. work authorized;

TECHNICAL SKILLS

AI / ML & LLM: LangChain, LangGraph, RAG, retrieval pipelines, agentic workflows, tool-calling, prompt engineering, DeepEval, Azure OpenAI, OpenAI Realtime, Gemini 2.0, function calling, structured outputs, guardrails, hallucination detection
Backend & APIs: Node.js, Python, tRPC, REST, async/streaming, multi-tenant systems, API-first design, Inngest, Next.js
Data & Databases: PostgreSQL, MongoDB, Prisma, SQL, data modeling, ETL/ELT pipelines, PySpark, Databricks, Amazon RDS
Infrastructure & Observability: Docker, Kubernetes, Azure, AWS (S3), GCP, CI/CD, OpenTelemetry, monitoring, MCP (Model Context Protocol)
Streaming & Voice: Twilio WebSockets, Deepgram STT, Stream Video SDK, WebRTC, real-time audio processing

PROJECTS

Brand Guardian AI - Video Compliance Auditing Platform | *LangGraph, Azure OpenAI, Azure AI Search, Azure Video Indexer, RAG*

- Engineered a multi-step LangGraph RAG pipeline that ingests brand compliance videos, extracts timestamped evidence via Azure Video Indexer, and grounds every finding in cited retrieval results—eliminating unattributed claims.
- Built grounded retrieval on Azure AI Search with citation tracking, enabling auditors to trace each compliance flag back to a specific video segment.
- Integrated DeepEval hallucination and citation-coverage metrics directly into the CI/CD pipeline, gating deployments on quality thresholds and preventing regressions before production.
- Shipped LLM guardrails as a deploy-time safety net, reducing hallucination rate in post-release audits by establishing a measurable baseline for every model update.

Elara - Real-Time AI Meeting Voice Agent | *OpenAI Realtime, Stream Video SDK, Inngest, Next.js, TypeScript, tRPC, PostgreSQL, OpenAI*

- Built a real-time voice agent combining OpenAI Realtime API and Stream Video SDK to deliver live speech-to-text summarization with sub-second transcription latency during active meetings.
- Designed Inngest-based async orchestration with idempotent retries and JSONL KPI event logging, making long-running inference jobs resilient to transient failures and providing a queryable quality-analytics trail.
- Architected a multi-tenant Next.js/tRPC/PostgreSQL backend supporting per-tenant meeting capture, async summarization, and isolated reporting dashboards—scalable across independent organizations.

Websites - Agentic Code Generation SaaS | *Next.js, Gemini 2.0, LLM Agents, Inngest, E2B, PostgreSQL, Prisma*

- Designed a tool-using agentic system on Gemini 2.0 that orchestrates multi-step code generation jobs; offloaded long-running tasks to Inngest with durable retries, eliminating mid-run failures on jobs exceeding LLM context windows.
- Implemented automated code validation inside E2B sandboxes—running generated code against test suites and serving live previews—before any output reached users, achieving a verifiable correctness gate.
- Enforced structured outputs and deterministic state transitions across all tool calls and workflow steps, preventing runaway agents, infinite loops, and unintended spend.
- Built usage metering and credit-based billing, enabling per-user consumption tracking and supporting tiered monetization at launch.

Real-Time Pharmacy Voice Agent | *Twilio WebSockets, Deepgram STT, Function Calling, Docker, Kubernetes*

- Built an end-to-end real-time pharmacy voice agent on Twilio WebSockets with Deepgram STT and LLM function-calling workflows to handle prescription inquiries without human handoff.
- Reduced end-to-end voice response latency to **1.5–2.5s** (p50/p95) through 20ms audio frame tuning and barge-in handling, meeting conversational UX requirements.
- Containerized with Docker and deployed on Kubernetes with rolling updates and safe rollbacks, supporting load spikes without downtime.
- Added call-level latency and failure-rate monitoring, enabling data-driven tuning of the STT and LLM pipeline post-launch.

EXPERIENCE

Software Engineer Intern — Backend Platform & Integrations

Jul 2023 – Dec 2023

Club InQuizitive

- Scaled the Node.js API platform from 0 to **1,000+ users** by containerizing v1/v2 services with Docker, enabling consistent deployments and reducing environment-related incidents across integrations.
- Diagnosed a p95 latency regression; profiled PostgreSQL query plans, rewrote 3 high-cost joins, and added targeted indexes—restoring response times to baseline and preventing further regressions.
- Collaborated with Product and external stakeholders to define platform workflow contracts, translating ambiguous requirements into documented API specifications adopted across two integration partners.

- Instrumented critical API paths with OpenTelemetry traces, metrics, and structured logs, providing the team its first end-to-end visibility into request performance and reducing mean time-to-detect issues.

Data Engineering Intern — Data Pipelines & Quality Sep 2022 – Feb 2023

JavaTpoint

- Built Python/SQL ETL pipelines to ingest, validate, and standardize schemas across multiple new datasets, enabling downstream teams to trust and consume data without manual cleaning.
- Improved ETL reliability by refactoring slow transforms, adding idempotent retries, and introducing unit tests—reducing pipeline failure rate and eliminating repeated re-runs during onboarding.
- Implemented checksum validation and structured run logs to enforce data lineage tracking, enabling the team to detect schema drift early and pinpoint root causes within minutes instead of days.

EDUCATION

Master of Science in Computer Science May 2026

University of North Texas *Denton, TX*

- Focus: AI/ML & Data Science | Relevant coursework: Distributed Systems, Databases, Data Engineering (30+ graduate credits)

Bachelor of Technology in Computer Science 2024

B V Raju Institute of Technology

CERTIFICATIONS

Machine Learning & AI (ML fundamentals, evaluation metrics, LLMs, prompt engineering, generative AI systems) — DeepLearning.AI

Building AI Agents & Responsible LLM Deployment — Anthropic